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Carbon on the Chain: Digital Solutions for a Greener Planet

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SUBMITTED BY

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CERTIFICATE

Certified that the project work entitled "**Carbon on the Chain: Digital Solutions for a Greener Planet**" carried out by **Chandana S bearing USN 1JB22IS032, Deekshitha R bearing USN 1JB22IS045, Divya Devaraj Hegde bearing USN 1JB22IS05, Ganashree B S bearing USN 1JB22IS062,** are bonafide students of SJB Institute of Technology in partial fulfillment for the award of BACHELOR OF ENGINEERING in INFORMATION SCIENCE AND ENGINEERING of the Visvesvaraya Technological University, Belagavi during the Academic Year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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Regards,

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ABSTRACT

As the global climate crisis intensifies, the urgency for sustainable solutions to reduce greenhouse gas emissions has never been greater. One of the most prominent mechanisms employed to incentivize emission reductions is the carbon credit trading system, particularly the cap-and-trade model. However, traditional implementations of carbon credit systems suffer from several critical limitations, including opacity, risk of fraud, double-counting of credits, limited scalability, and inefficiencies in regulatory compliance and verification processes. This project proposes an innovative blockchain-powered carbon credit ecosystem that addresses these inherent flaws by leveraging the core strengths of distributed ledger technology—transparency, immutability, decentralization, and automation.

The system enables the creation, monitoring,trading, and verification of tokenized carbon credits using a blockchain infrastructure, primarily built on Ethereum and supported by smart contracts coded in Solidity. It incorporates a decentralized cap-and-trade mechanism, where carbon emissions are capped and tradable permits (carbon credits) are allocated, creating a financial incentive for emitters to reduce their carbon output. To ensure trust and accountability, the solution integrates a Monitoring, Reporting, and Validation (MRV) framework directly onto the blockchain, thereby enabling real-time and tamper-proof recording of emission data. Smart contracts automate critical processes such as credit issuance, compliance enforcement, and transaction settlement, reducing the need for manual intervention and third-party verification.

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CHAPTER-1

INTRODUCTION

1.1 OVERVIEW

Carbon on the Chain: Digital Solutions for a Greener Planet is an innovative blockchain-based platform designed to tackle the longstanding inefficiencies in traditional carbon markets, such as fraud, double counting, opaque verification processes, and administrative delays. Traditional carbon credit systems often rely on centralized authorities and manual auditing, which creates delays, reduces trust, and limits participation, especially for smaller organizations. By integrating Ethereum blockchain, smart contracts, NFT-based registration, and IPFS for decentralized storage, Carbon on the Chain ensures secure registration of organizations, tamper-proof emission data storage, and automated issuance of carbon credits, effectively removing human error and the risk of data manipulation.

Under the Cap-and-Trade framework, organizations first register on the platform and receive NFT-based certificates upon successful verification by authorized auditors. Verified emission reductions are converted into ERC-20 carbon credit tokens that can be traded on a decentralized Power Exchange Portal using a double-auction mechanism. This mechanism guarantees transparent and fair price discovery while allowing real-time, trustless trading. By decentralizing registration, MRV (Monitoring, Reporting, Verification), and trading, the platform ensures accountability at every stage of the carbon credit lifecycle. All emission data, audit reports, and transaction histories are stored immutably on the blockchain and IPFS, providing an auditable and verifiable trail that regulators, auditors, and stakeholders can access securely.

Beyond ensuring operational efficiency and transparency, the platform actively encourages sustainable practices by rewarding organizations for reducing emissions below their allocated cap. It also facilitates broader participation in the carbon market by providing an accessible, user-friendly interface built on the MERN stack with Web3 and MetaMask integration. The ecosystem is designed to scale in the future to support IoT-based emission tracking, integration with national and international carbon registries, and cross-border carbon trading, enabling a globally interoperable solution. By combining blockchain technology with smart contracts, tokenization, and decentralized storage, Carbon on the Chain provides a modern, reliable, and secure framework for carbon credit management, helping industries, governments, and auditors contribute more effectively to global climate action and sustainability goals.

In essence, Carbon on the Chain is not only a technological solution but also a strategic enabler for transparent, efficient, and accountable carbon markets. It represents a shift from centralized, manual carbon trading systems to a digital-first, automated, and decentralized ecosystem that strengthens trust, incentivizes emission reductions, and supports nations in achieving their environmental commitments under global agreements such as the Paris Accord.

The project utilizes a combination of modern development tools, programming languages, and frameworks to build a secure and efficient blockchain-based carbon credit ecosystem. Smart contracts are developed using Solidity and deployed on Ethereum-compatible networks through Remix IDE**,** supported by Ganache/Hardhat for local blockchain simulation. MetaMask is used for wallet authentication and transaction signing, The user interfaces and dashboards are built using HTML, CSS, and JavaScript**,** with Lucide Icons enhancing visual components and Chart.js providing analytical data visualizations. jsPDF is used to generate secure, downloadable certificate PDFs. Together, these tools form a robust full-stack environment that supports registration, verification, token minting, and carbon-credit trading on a decentralized platform.

1.2 SCOPE OF PROJECT

The scope of this project encompasses the design, development, and demonstration of a comprehensive blockchain-powered carbon credit trading ecosystem aimed at enhancing transparency, security, and automation across the entire carbon credit lifecycle. The platform covers all critical stages of carbon credit management, including registration, monitoring, verification, issuance, and trading, ensuring a seamless and trustless workflow for all participants.

A key component of the system is the decentralized registration module, where organizations receive NFT-based certificates that serve as verifiable digital identities on the blockchain. This ensures that all participating entities are authenticated, reducing the potential for fraudulent activity.

The project also implements a structured MRV (Monitoring, Reporting, Verification) workflow, enabling auditors to validate emission data securely. These validated reductions are automatically converted into ERC-20 carbon credit tokens through smart contracts, eliminating manual intervention and ensuring immediate issuance of tokens based on verified data.

The project further includes the development of a decentralized Power Exchange Platform that allows organizations to trade carbon credits in a secure, transparent manner. A double-auction mechanism ensures fair price discovery, facilitating equitable market participation for both small and large organizations. To ensure data integrity and immutability, the platform incorporates IPFS (InterPlanetary File System) for tamper-proof document storage, while Web3 and MetaMask integration provides a seamless interface for blockchain interactions. A MERN-based web application offers user-friendly access, dashboards, and management tools for organizations, auditors, and regulators.

While the current prototype demonstrates the feasibility and effectiveness of blockchain in improving transparency, preventing manipulation, and reducing fraud in carbon markets, the system is limited to simulated emission data, test networks, and pilot-scale MRV processes. Nonetheless, the architecture is designed to support future scalability, including integration with IoT-enabled emission sensors for real-time monitoring, expansion to large-scale industrial deployment, and interoperability with national and international carbon registries. The scope also envisions enhancements such as automated compliance reporting, predictive analytics for emission trends, and cross-border trading, laying the groundwork for a fully decentralized, global carbon credit ecosystem.

Overall, this project provides a robust foundation for modernizing carbon markets, offering a secure, efficient, and transparent system that can evolve into a scalable solution for real-world carbon management, regulatory compliance, and climate action initiatives.

1.3 MOTIVATION

Climate change is one of the most urgent challenges facing the world today, with far-reaching environmental, social, and economic consequences. Despite the critical role of carbon credit systems in mitigating greenhouse gas emissions, traditional carbon markets continue to operate with inefficiencies that limit their overall effectiveness.

These inefficiencies include lack of transparency, centralized control, slow and manual verification processes, fraudulent activities, double counting of credits, and limited accessibility for smaller organizations. Such challenges not only undermine trust among industries, governments, and auditors but also weaken the global effort to reduce carbon emissions.

As countries like India and others commit to ambitious emission reduction targets under international agreements such as the Paris Accord, there is a growing need for modern, reliable, and tamper-proof mechanisms that can effectively track, verify, and facilitate the trading of carbon credits. Current systems, reliant on paper-based or semi-digital processes, are unable to provide real-time tracking, automated verification, or secure record-keeping. These limitations slow down the credit lifecycle, create opportunities for manipulation, and discourage smaller organizations from participating in carbon markets, thereby limiting the reach and impact of climate initiatives.

Blockchain technology, with its decentralized, immutable, and transparent architecture, offers a transformative solution to these challenges. By leveraging blockchain, smart contracts, NFTs, and decentralized storage, the proposed system enables secure registration of organizations, automated verification of emission reductions, and instantaneous issuance of carbon credits. This approach ensures a verifiable and auditable record of every transaction, enhancing trust and accountability among all stakeholders.

The motivation for this project extends beyond technological innovation; it is driven by the need to foster sustainability and promote greener industrial practices. By providing a transparent and efficient carbon credit ecosystem, organizations are incentivized to reduce their carbon footprint and actively participate in climate action initiatives. Furthermore, such a system can support national and international climate policies, facilitate compliance reporting, and pave the way for large-scale adoption of automated carbon markets. Ultimately, the project demonstrates how emerging technologies can modernize environmental governance, strengthen global climate action, and contribute to a sustainable and accountable future where carbon markets are more transparent, reliable, and inclusive for all stakeholders.

1.4 OBJECTIVES

The primary objective of this project is to develop a transparent, secure, and efficient carbon credit trading system leveraging blockchain technology to eliminate fraud, manipulation, and double counting. The system aims to automate the entire carbon credit lifecycle, including registration, emission reporting, verification, credit issuance, and trading, through smart contracts, reducing manual intervention and enhancing trust among stakeholders. NFT-based registration certificates will ensure authentic and tamper-proof identification of organizations participating in the carbon market, while decentralized storage via IPFS and blockchain-based MRV processes will maintain immutable and verifiable emission records.

The project also seeks to design a decentralized Power Exchange Platform that facilitates fair and efficient trading of carbon credits using a double-auction mechanism, and to simplify verification by enabling auditors to validate emission data digitally and store results securely on-chain. Beyond technological implementation, the system encourages sustainability by incentivizing organizations to reduce emissions below their allotted cap and adopt greener industrial practices.

A user-friendly web interface built using the MERN stack, integrated with Web3 and MetaMask, ensures seamless interaction with the blockchain, making the platform accessible to all participants. Finally, the project aims to demonstrate a functional prototype that can be scaled in the future for real-world deployment, integration with national and international carbon registries, and IoT-enabled emission tracking, thereby providing a comprehensive, reliable, and modern solution for carbon credit management.

CHAPTER 2

LITERATURE SURVEY

The objective of carrying out literature survey before implementing any project is to get an idea of how the project can be implemented and to understand various technologies used to build a project. So it is of utmost necessary to find the most recent and promising research papers related to our project topic "Carbon Credit Ecosystem Powered by Blockchain Technology". The papers referred to have their publication dates in range between 2020 and 2025. The details about the contributions made by each author are given below.

2.1 Research Papers

Some of the papers referred are as follows:

Application of Blockchain Technology in Carbon Trading Market: A Systematic Review-2025.

The review also highlights how, blockchain can transform carbon trading by creating globally connected markets through interoperable digital infrastructures. Standardized blockchain-based data formats can harmonize emission reporting across countries, reducing inconsistencies in cross-border carbon credit transfers. The study compares governance models such as DAOs, permissioned blockchains, and hybrid structures, noting that each offers varying levels of transparency, regulatory control, and scalability. Smart contracts are identified as key enablers for automating credit issuance, enforcing emission caps, allocating penalties, and managing credit expiry, thereby reducing manual intervention and supporting real-time policy enforcement. Blockchain's immutable records also make auditing faster and more reliable, lowering verification costs and increasing trust in carbon offset projects. The review further addresses challenges such as the digital divide, emphasizing the need for infrastructure support and capacity building in developing regions. Overall, the study concludes that blockchain can significantly enhance transparency, traceability, fraud prevention, and global coordination in carbon markets, but achieving this requires strong regulations, improved scalability, and international collaboration.

Blockchain-Powered Carbon Credit Trading System Using Cap-and-Trade Mechanism-2024.

The study further emphasizes that the blockchain-powered cap-and-trade system can greatly enhance Monitoring, Reporting, and Verification (MRV) by integrating IoT sensors and automated data feeds, enabling real-time and tamper-proof emission tracking. Smart contracts can instantly deduct credits or apply penalties when thresholds are exceeded, ensuring strict compliance and reducing manual audits. The research also highlights that blockchain provides a unified, shared ledger for regulators, industries, auditors, and traders, eliminating data inconsistencies and lowering administrative overhead while improving system resilience.

The study notes that dynamic, transparent pricing of credits and smart contract–based auctions can improve market liquidity and encourage proactive emission reduction. It also emphasizes the system's potential for international carbon trading by creating standardized digital carbon assets that support cross-border exchanges aligned with the Paris Agreement's Article 6 mechanisms. Challenges remain, including the need for strong digital infrastructure, regulatory alignment, user training, data governance, and scalable blockchain networks. Overall, the paper concludes that combining immutability, automation, secure trading, and real-time monitoring makes blockchain a powerful tool for strengthening future global cap-and-trade systems.

Blockchain-Powered Carbon Credit Management: Innovating Sustainability Tracking-2024.

This paper focuses on blockchain can modernize carbon credit management systems that currently struggle with inaccuracies, double counting, slow verification, and dependence on centralized authorities. It proposes a decentralized model that records the entire lifecycle of carbon credits—from issuance and validation to trading and retirement—on an immutable ledger. Smart contracts automate verification, auditing, and transaction settlement, reducing human error and lowering administrative costs. The system also supports integration with third-party verification agencies and introduces standardized digital carbon certificates to improve global interoperability.Overall, the paper concludes that blockchain can unify fragmented carbon registries and build a trusted, globally compatible sustainability tracking ecosystem.

A Low-Code Platform for Carbon Credit Trading Using Blockchain Technology-2024.

This study presents a comprehensive exploration of a blockchain can improve carbon credit management systems that face issues like inaccuracies, double counting, slow verification, and reliance on centralized authorities. By using a decentralized and immutable ledger, the system securely tracks the entire carbon credit lifecycle—from issuance to trading and retirement. Smart contracts automate verification, auditing, and settlement, reducing human error and administrative effort. The model also enables integration with third-party verifiers and supports standardized digital carbon certificates for global use. While the approach offers benefits such as faster credit issuance, greater transparency, and lower operational costs, it also faces challenges including data integration, scalability, and regulatory compatibility. Overall, the study concludes that blockchain can unify fragmented carbon registries and create a more reliable and globally interoperable sustainability tracking system.

Empowering India’s Climate Action: Harnessing Blockchain for Carbon Trading-2024.

This paper explores the potential of blockchain technology to strengthen India’s emerging carbon trading ecosystem. Despite India’s commitment to reducing emissions, existing carbon systems face challenges such as fraudulent credit reporting, lack of trust among stakeholders, unclear regulatory mechanisms, and inefficient data management. The study proposes a hybrid blockchain model that stores sensitive data off-chain for scalability and non-sensitive transaction data on-chain for immutability and transparency. This dual-layer approach optimizes system performance while maintaining data integrity. The blockchain-enabled platform ensures secure carbon credit issuance, real-time tracking, and transparent trading, supporting India's climate policy frameworks and industrial sustainability targets. The study also discusses how blockchain can facilitate compliance with international agreements and create a unified carbon registry for the country. However, the research also identifies barriers such as lack of digital literacy, regulatory gaps, technology adoption costs, and need for cross-sector collaboration. The authors emphasize that blockchain could significantly modernize India’s carbon market while accelerating national climate action

Transparency in Carbon Credit by Automating Data Management Using Blockchain-2022.

The study explores that blockchain can greatly improve carbon credit management by replacing manual, centralized processes with a transparent and tamper-proof automated system. Traditional frameworks often suffer from delays, inconsistent reporting, and risks of fraud, whereas a blockchain-based approach unifies emission recording, verification, credit issuance, trading, and retirement into a single decentralized ecosystem. Smart contracts ensure that credits are issued only after verified emission reductions, reducing human error and manipulation.Integrating blockchain with IoT sensors enables continuous monitoring and automated MRV, making the process more accurate and efficient. Although challenges such as high implementation costs, legacy system integration, and regulatory differences exist, the study notes that standardized protocols and policy support can address these issues.

Sustainable Production Inventory Model with Green Technology Investment Under Carbon Tax and Cap-and-Trade Policy-2024.

This research develops presents an Economic Production Quantity (EPQ) model that incorporates carbon tax and cap-and-trade policies while analyzing how green technology investments affect production and inventory decisions. It shows that environmental regulations significantly shape manufacturing costs, strategies, and profitability. Carbon taxes provide stable and predictable pricing, whereas cap-and-trade costs fluctuate with market conditions. Although green technologies require high initial investment, they lower emissions and operational costs over time. Sensitivity analysis reveals how changes in carbon prices, production rates, and demand impact profitability, helping firms plan effectively under different policy environments. The study also compares outcomes across regulatory frameworks, offering useful insights for policymakers.

Carbon Emission Monitoring and Credit Trading: The Blockchain and IoT Approach-2021.

This study presents a framework that integrates IoT sensors with blockchain to improve carbon emission monitoring and carbon credit trading. IoT devices continuously capture real-time emission data, offering more accurate and detailed insights than periodic manual inspections. This data is securely recorded on the FISCO BCOS blockchain, ensuring immutability, traceability, and protection against tampering. Smart contracts automate carbon credit issuance, apply penalties for noncompliance, and enable secure, auditable trading. The combined system enhances transparency, accuracy, and efficiency, strengthening trust among regulators, auditors, and industries. Performance results show faster verification, fewer transaction errors, and better responsiveness, demonstrating its practical viability.Overall, research highlights this IoT-blockchain approach as a scalable,secure solution that streamlines compliance, supports regulatory enforcement, and enables continuous environmental monitoring for modern carbon markets.

Carbon Credits on Blockchain-2020.

This study proposes a blockchain-based tokenization model for carbon credits, where emission permits are converted into digital tokens that can be securely traded on decentralized networks. This approach enhances transparency, traceability, and market liquidity while preventing double-spending and unauthorized credit creation. Smart contracts and blockchain consensus mechanisms ensure secure transactions, tamper-proof records, and faster settlements compared to traditional systems. The model also opens opportunities for smaller producers to participate in carbon markets by reducing administrative barriers. The study discusses key technical aspects such as smart contract design, security, governance, and interoperability for cross-border trading. It also highlights regulatory challenges, stressing the need for global policy alignment and standard protocols.

Energy Sector and Carbon Trading in the Future: A Reference to India-2024

This study reviews the future of carbon trading in the energy sector with a focus on India’s role in global climate agreements. It highlights India’s growing contribution to Certified Emission Reductions (CERs) and examines how national policies on renewable energy and industrial emissions align with international carbon markets. The study notes that India’s industrial growth and renewable capacity create strong potential for expanding carbon trading while supporting sustainable development. It also identifies challenges such as market volatility, inconsistent pricing, regulatory uncertainty, and low stakeholder awareness. Overall, the research concludes that stronger carbon market participation can help India meet climate goals, boost economic growth, and support the global transition to low-carbon energy systems.

Energy Sector and Carbon Trading in the Future: A Reference to India – Bhardwaj M.

Bhardwaj explores the evolving role of carbon trading as a strategy to mitigate global warming, with a specific focus on India's position in international climate policy. The study highlights how agreements such as the Kyoto Protocol and the Paris Agreement encouraged the formation of global carbon markets by enabling the trade of greenhouse gas (GHG) reduction units. It examines carbon credit allocation mechanisms, market-driven pricing, and India's significant participation through renewable energy projects and Clean Development Mechanism (CDM) initiatives.

The paper concludes that India possesses strong potential to leverage carbon trading for sustainable economic expansion, provided regulatory frameworks are strengthened, transparency is enhanced, and advanced digital tools are adopted to prevent fraud and streamline monitoring.

Transforming Carbon Emission Trading Schemes in Thailand Using Non-Fungible Tokens

The study further highlights how integrating NFTs into Thailand's carbon emission trading scheme can modernize the nation's climate governance infrastructure. By leveraging blockchain's immutable ledger and NFT standards, the proposed framework ensures that every carbon credit issued, traded, or retired is cryptographically secured and permanently recorded. This eliminates common issues such as double counting, unauthorized credit alterations, and opaque audit trails that often undermine trust in traditional carbon markets.

Moreover, the system establishes a decentralized, real-time monitoring and verification mechanism that significantly reduces the reliance on manual documentation and third-party intermediaries. Smart contracts automate compliance checks, issuance of verified credits, and settlement of trades, thereby shortening processing times and reducing administrative costs for both regulators and industry participants. The use of NFTs also supports enhanced interoperability, enabling cross-platform trading and potential regional integration with ASEAN carbon markets in the future.

In addition, the study emphasizes the potential socioeconomic benefits of this transformation. By creating a more transparent and efficient carbon trading environment, the NFT-based model encourages greater participation from industries, small businesses, and renewable energy producers. It also fosters investor confidence, attracts green financing, and aligns Thailand with global digital-asset-driven climate initiatives. Overall, the adoption of NFT-backed carbon credits presents a scalable, secure, and future-ready framework that could serve as a blueprint for other developing nations aiming to reform their carbon market ecosystems.

Resident Carbon Credit Incentive Decision-Making Method to Promote Valley Filling

The study further extends the incentive framework by integrating behavioral modeling to understand the responsiveness of residential consumers to carbon-credit rewards. By analyzing historical electricity consumption datasets, the authors apply K-means clustering to classify users into distinct behavioral groups—such as high-consumption households, flexible-load users, and time-sensitive consumers. This segmentation allows the development of personalized incentive packages that better match resident lifestyles and maximize participation in valley-filling programs.

To ensure optimal distribution of incentives, Grey Wolf Optimization (GWO) is utilized as an intelligent optimization technique capable of handling nonlinear, multi-objective constraints. GWO identifies ideal carbon-credit reward levels that balance grid stability, resident comfort, and economic feasibility. By running multi-scenario simulations, the researchers demonstrate that optimized incentives not only flatten peak loads but also significantly improve overall load factor and reduce carbon intensity during high-demand hours.

The paper also highlights the broader implications of the proposed methodology. The introduction of carbon-credit-based rewards fosters a consumer-centric energy ecosystem where residents are directly motivated to engage in sustainable energy usage. This enhances grid reliability, supports renewable energy integration, and reduces the operational costs associated with peak-hour electricity generation. Furthermore, the framework offers policymakers a robust decision-support tool, enabling them to design dynamic pricing schemes, promote carbon-neutral homes, and drive national decarbonization strategies. Overall, the methodology provides a scalable and data-driven model for future green electricity markets.

Carbon Credit Transfer System using Blockchain

The study further emphasizes how the integration of blockchain technology fundamentally transforms the carbon credit transfer lifecycle. By utilizing decentralized smart contracts, the proposed framework enforces strict rules for credit creation, retirement, and transfer, ensuring that no unauthorized or duplicate credits enter the system. Each carbon credit is tokenized as a unique digital asset, allowing transparent provenance tracking from issuance to final consumption. This enhances trust among stakeholders by making all transactions instantly verifiable and tamper-proof.

In addition, the system employs an Automated Market Maker (AMM) model to establish a decentralized trading environment where carbon credits can be exchanged without traditional intermediaries. The AMM ensures continuous liquidity by algorithmically determining prices based on supply–demand dynamics, thereby preventing bottlenecks commonly seen in manually operated or regulated exchanges. This decentralized exchange mechanism enables faster settlement times, reduces the risk of price manipulation, and promotes broader participation from industries, consumers, and environmental organizations.

Overall, the blockchain-enabled carbon credit transfer system represents a robust, scalable, and secure alternative to conventional mechanisms. By ensuring transparency, liquidity, and integrity, the proposed model supports global efforts to enhance the credibility and effectiveness of carbon markets.

Carbon-Credit Monitoring and Prediction in Smart Factory using Explainable AI and Data Analytics

This research develops an AI-driven carbon credit monitoring and prediction system tailored for smart factories. Machine learning models, particularly Bagged Tree classifiers, are used to predict-emissions.

Explainability is achieved using SHAP (Shapley Additive Explanations), which helps interpret the influence of individual parameters such as machine load, energy usage, and operational patterns.The system enables real-time insights, transparent decision-making, and improved emission forecasting, thus supporting factories in meeting carbon compliance. It bridges AI, analytics, and carbon credit mechanisms in Industry 4.0 environments.

Case Study of Carbon Accounting Information Disclosure Under the Background of Green Credit (AHP-Based)

This study assesses carbon accounting disclosure practices among 10 coal-listed companies using the Analytic Hierarchy Process (AHP). Four major dimensions—emission reduction, carbon-related activities, low-carbon strategy, and carbon-reduction action plans—form the evaluation-model.

Results reveal disparities in disclosure levels, with several companies lacking transparency in reporting emissions, mitigation strategies, and carbon performance data.

The study emphasizes the importance of comprehensive and standardized disclosure to gain green credit benefits, improve corporate accountability, and support environmental governance.

A Study on the Application of Blockchain in Carbon Trading Systems

This paper provides a detailed analysis of blockchain's applicability in carbon trading markets. It categorizes existing architectures and highlights blockchain's core advantages: decentralized verification, immutable record-keeping, fast settlement, and fraud prevention.

The authors discuss both technical and regulatory challenges faced by blockchain-based carbon platforms, such as scalability limits, cross-border compliance, and integration with existing MRV(Monitoring,Reporting,Verification)structures.

The study concludes that hybrid on-chain/off-chain designs offer the best balance of transparency and cost efficiency and recommends future research into interoperable global carbon registries.

Blockchain-Based Energy Management System: A Proposed Model

This proposed model integrates blockchain with traditional and renewable energy systems to improve transparency and trust in energy transactions. Smart contracts govern energy distribution, pricing, and consumption records in a decentralized manner.

The model incentivizes sustainable behavior by rewarding green energy usage and supports peer-to-peer(P2P)energy-trading.

The authors highlight how blockchain can enhance energy reliability, support demandresponse programs, and establish secure, tamper-proof energy audits, making it a vital tool for next-generation smart grids.

Carbon Credits Storage: A Comparative Multifactor Analysis of On-Chain vs Off-Chain Solutions

The study further expands the comparative assessment by introducing a structured multifactor decision matrix that quantifies the performance trade-offs between on-chain and off-chain carbon credit storage architectures. The analysis considers not only technical parameters—such as transaction throughput, cryptographic security, and system latency—but also institutional factors, including regulatory compliance, auditability, and long-term data preservation. Through this multifaceted evaluation, the authors emphasize that no single storage method universally outperforms the other; instead, optimal design depends on the specific requirements of carbon market stakeholders.

In evaluating on-chain models, the paper highlights the benefits of fully decentralized verification, immutable data trails, and the elimination of intermediary-controlled registries. These strengths make on-chain storage highly suitable for environments where trust is limited and transparency is paramount. However, the study also notes significant drawbacks such as high gas fees, large carbon footprints associated with certain blockchains, and constraints on storing large files or frequently updated datasets directly on-chain.

Overall, the paper concludes that hybrid storage architectures offer a robust and future-ready framework capable of supporting global carbon markets as they transition toward digital, tokenized, and blockchain-integrated ecosystems.

Imperfect Production–Inventory Models with Carbon Cap-and-Trade and ACC Payment

This study extends traditional production–inventory models by incorporating deterioration effects, imperfect production, and carbon cap-and-trade policies. The framework considers raw material decay, defective outputs, inspection errors, and emission-related costs to better reflect real manufacturing environments dealing with perishable items. A key contribution is the integration of the advance-cash-credit (ACC) payment scheme, which helps firms improve cash flow, reduce financial pressure, and support cleaner production strategies.Results from the optimized model show that firms can enhance profits and simultaneously reduce carbon emissions by selecting appropriate replenishment cycles, controlling production quality, and effectively using ACC financing. Sensitivity analyses indicate that carbon trading prices strongly influence batch size, emission-control investments, and overall production decisions. Higher carbon prices push firms to adopt more efficient, low-emission operations, whereas lower prices favor cost-driven production.

Overall, the work provides practical insights for managers on balancing sustainability goals with profitability, demonstrating how integrated operational and financial strategies can improve both economic and environmental performance.

Blockchain-Based Credit System to Increase the Creditability through Carbon Offset Level

This study introduces a blockchain-based credit verification system designed to accurately track carbon offsets in the agricultural sector. By recording offset activities—such as soil carbon sequestration, organic cultivation, and renewable energy usage—on an immutable blockchain ledger, the platform prevents data tampering, double counting, and fraudulent sustainability claims. Each carbon contribution is tokenized, enabling transparent monitoring across the supply chain.The system's integration of Augmented Reality (AR) provides an interactive layer for consumers. By scanning product labels, users can instantly visualize carbon offset levels, trace farm origins, and view certification details in a user-friendly AR interface. This not only enhances product transparency but also increases consumer confidence in sustainability claims.

For farmers, the platform ensures fair compensation by converting verified carbon offsets into tradable digital credits, creating new income opportunities for adopting green practices. Retailers and regulators also benefit from streamlined verification processes and reduced dependence on manual documentation.Overall, this hybrid AR–Blockchain approach strengthens the credibility of carbon-neutral agricultural products, improves traceability, and encourages broader adoption of environmentally responsible farming, thereby contributing to long-term sustainable development.

2.2 Existing System

The following are the drawbacks found in the existing system:

- Risk of fraud, manipulation, and double counting across registries.
- Lack of transparency in carbon credit generation, reporting, and trading.
- Slow and complex verification (MRV) processes, causing delays in credit issuance.
- Centralized systems are prone to corruption and single-point failure.
- Integration challenges between new technologies and existing legacy systems.

2.3 Proposed System

In the proposed system we aim to

1. Lack of transparency in carbon credit generation, reporting, and trading
   - Blockchain + Smart contracts + NFT certificates provide transparency.
   - IPFS storage provides transparent reporting.
2. Risk of fraud, manipulation, and double counting across registries
   - Smart contracts automate issuance and prevent fraud.
   - NFT-based certificates prevent duplication.
3. Slow and complex MRV (Monitoring, Reporting, Verification) processes
   - MRV automation is mentioned in abstract.
4. Centralized systems prone to corruption & single-point failures
   - Uses decentralized blockchain + IPFS → eliminates central dependency.
5. Most solutions are prototypes, lacking real deployment
   - Project also as a prototype (Etherum + Ganache + basic MRV + report storage).
   - Pending features show it is still not fully production-ready.
6. certification records
   - verified carbon emissions of an organization,including total CO_2 output and Scope 1, 2, and 3 values.
   - It confirms data accuracy through MRV-based verification and audit validation.

CHAPTER-3

SYSTEM REQUIREMENTS

3.1 Software Requirements

The project requires the following software components for development, deployment, smart contract execution, blockchain connectivity, and user interface rendering:

1. Blockchain & Smart Contract Software

- Remix IDE – For writing, compiling, and deploying Solidity smart contracts
- Ganache / Hardhat – Local blockchain simulation and testing
- MetaMask Wallet – Wallet for authentication, signing transactions, and interacting with the blockchain
- Ethereum / Polygon Testnet – Execution environment for smart contracts

2. Backend Software

- Node.js – Server-side runtime environment
- Express.js – Backend framework for API creation
- Web3.js / Ethers.js – Connects backend & frontend to blockchain

3. Frontend Software

- HTML, CSS, JavaScript – User interface structure and styling
- React.js / Next.js – Dynamic and interactive dashboards

4. Certificate & Report Generation

- jsPDF – For generating downloadable PDF certificates (Valid / Invalid)
- Chart.js – For graphical analytics and emission visualization

5. Development & Testing Tools

Visual Studio Code (VS Code) – Primary code editor

6. Browser Requirements

- Google Chrome / Edge – Compatible browsers
- MetaMask Extension Installed – Mandatory for blockchain transactions

3.2 Hardware Requirements

Minimum Requirements

- Intel Core i5 processor (or equivalent)
- 8 GB RAM
- 20 GB free disk space
- Integrated graphics (GPU optional)
- Stable internet connection
- Standard keyboard and mouse

Recommended Requirements

- Intel Core i7 / AMD Ryzen 7 processor
- 16 GB RAM for smoother performance
- 50 GB SSD storage for faster read/write
- High-speed broadband internet
- 15.6-inch or larger display

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system architecture is designed as a modular, decentralized, and role-driven framework operating on the Ethereum blockchain, ensuring transparency, immutability, and secure participation. It consists of four integrated layers that collectively manage the end-to-end carbon credit lifecycle.

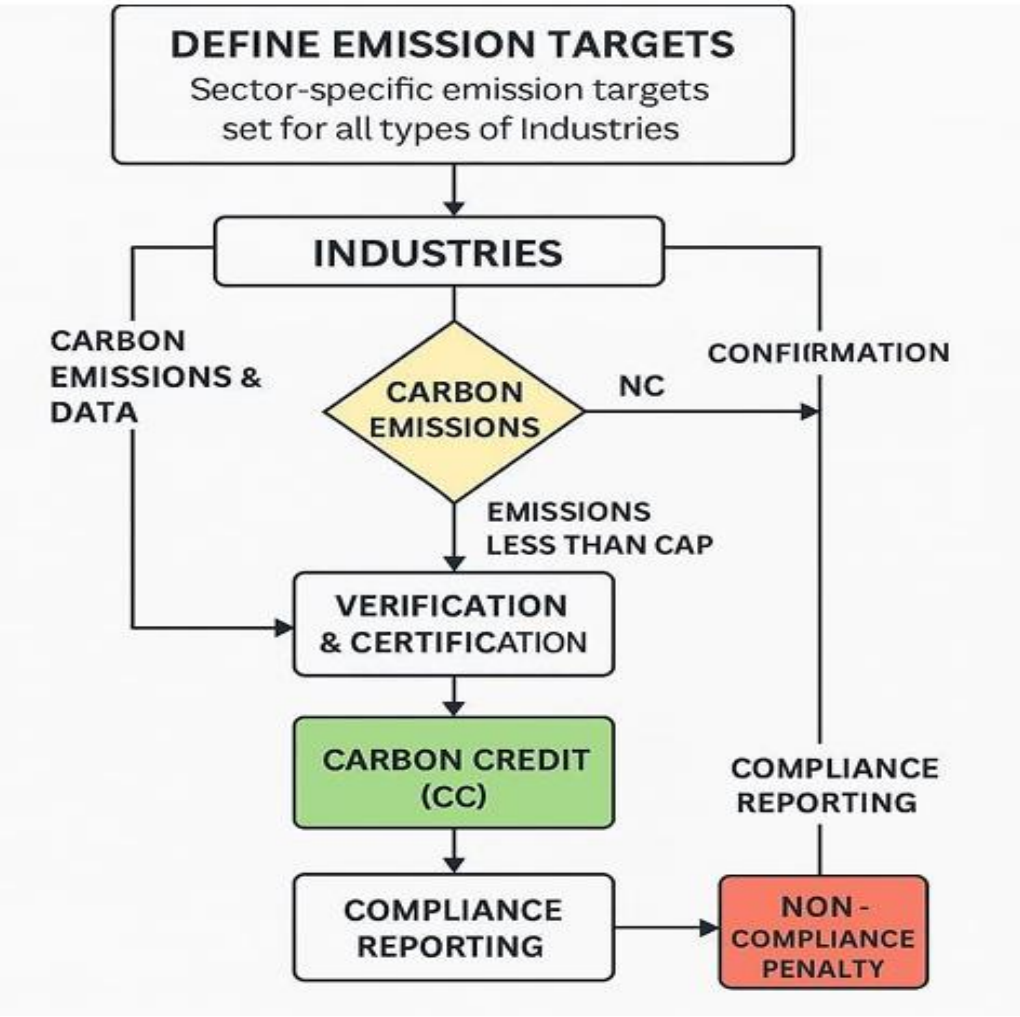

Fig 4.1: System Architecture

The Frontend Layer provides interactive, role-specific dashboards for POSOCO, BEE, auditors, organizations, and the trading exchange, built using the MERN stack and connected to Web3 for seamless blockchain interactions, wallet-based authentication, and NFT identity verification. The Smart Contract Layer forms the core logic, consisting of the ERC-20 carbon credit token contract, the ERC-721 NFT identity certificate contract, and a comprehensive Carbon Registry contract that governs role management, emission reporting, MRV verification, credit issuance, and trading controls. Beneath this, the Blockchain Layer ensures decentralized execution and immutability using Ethereum-compatible networks such as Ganache, Infura, or any EVM-compatible chain, maintaining a secure environment for on-chain operations.

Complementing these components, the Storage Layer uses a hybrid on-chain/off-chain approach, storing essential metadata and state transitions on the blockchain while leveraging IPFS and PDF generation for storing large emission reports, audit documents, and verification certificates, thus optimizing cost efficiency and data permanence. This layered architecture collectively enables a transparent, tamper-proof, and fully automated carbon credit management ecosystem.

4.2 Context Analysis Diagram

The Context Analysis Diagram represents the entire blockchain-based carbon credit system as a single high-level process and shows how all external entities interact with it. The central system—Carbon on the Chain Platform—receives inputs from multiple stakeholders and produces outputs through secure, blockchain-verified workflows.

4.2.1 Class Diagram

The class diagram represents the structural blueprint of the carbon credit management system by illustrating the key entities, their attributes, and the relationships between them. The CarbonEmitter class models organizations participating in the platform, storing essential details such as project information, emission reports, and status updates, while also providing functions to submit reports and request verification. The Regulator class represents governing authorities responsible for overseeing project evaluation, validating emission reductions, and approving or rejecting credit requests. The CarbonCredit class manages the issuance and tracking of carbon credits, linking each credit to the corresponding emitter and project, and maintaining attributes such as credit amount, token ID, and transaction history. Relationships among these classes reflect real-world interactions—emitters initiate and implement projects, regulators verify these projects, and validated reductions are converted into carbon credits. Together, the classes form a cohesive structure that supports transparency, automation, and traceability within the blockchain-based carbon credit ecosystem.

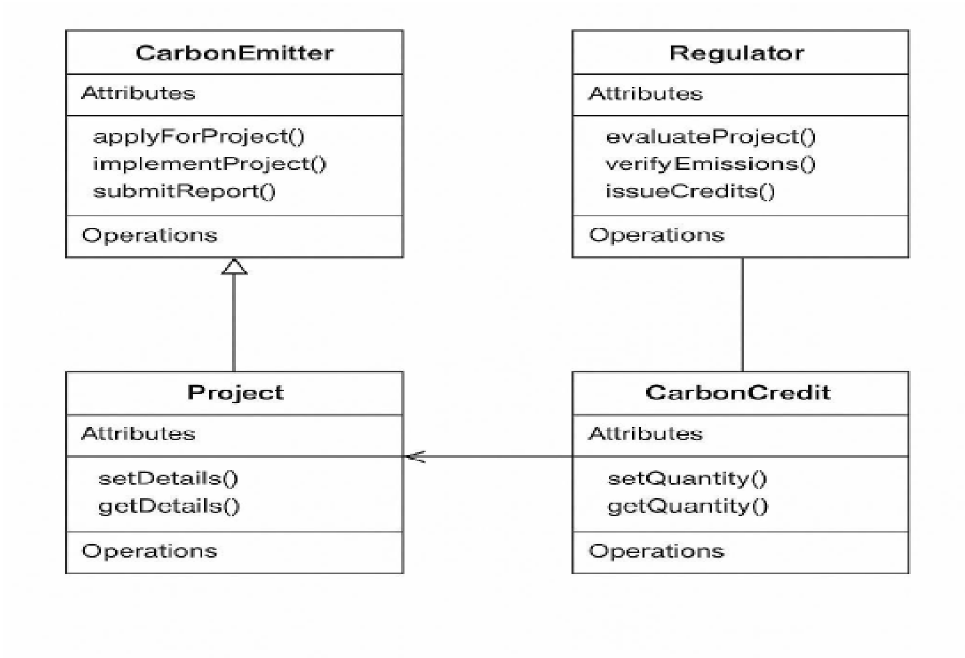

Figure 4.2.1: Class Diagram

4.2.2 Use Case Diagram

The use-case diagram illustrates how all key stakeholders interact with the Carbon on the Chain platform to ensure a transparent and automated carbon credit lifecycle. The process begins with the Emitter Organization**,** which registers on the system, obtains digital identity credentials, and submits its periodic emission data along with supporting documents. Once the data is uploaded, the Verifier or Auditor reviews the submitted reports, validates the authenticity of the emission measurements, and either approves or rejects the claims. These validation actions trigger predefined smart contract functions that record the outcome securely on the blockchain. After successful verification, the Regulator oversees compliance requirements and authorizes the issuance of carbon credits based on validated emission reductions.

The Blockchain Network then mints and transfers the corresponding ERC-20 carbon credit tokens to the emitter's wallet, ensuring tamper-proof tracking, auditability, and prevention of double counting. The use-case diagram captures this end-to-end workflow—registration, data submission, validation, approval, and credit issuance—demonstrating how each actor collaborates within a decentralized environment to maintain trust, automate processes, and ensure the integrity of carbon credit generation and management.

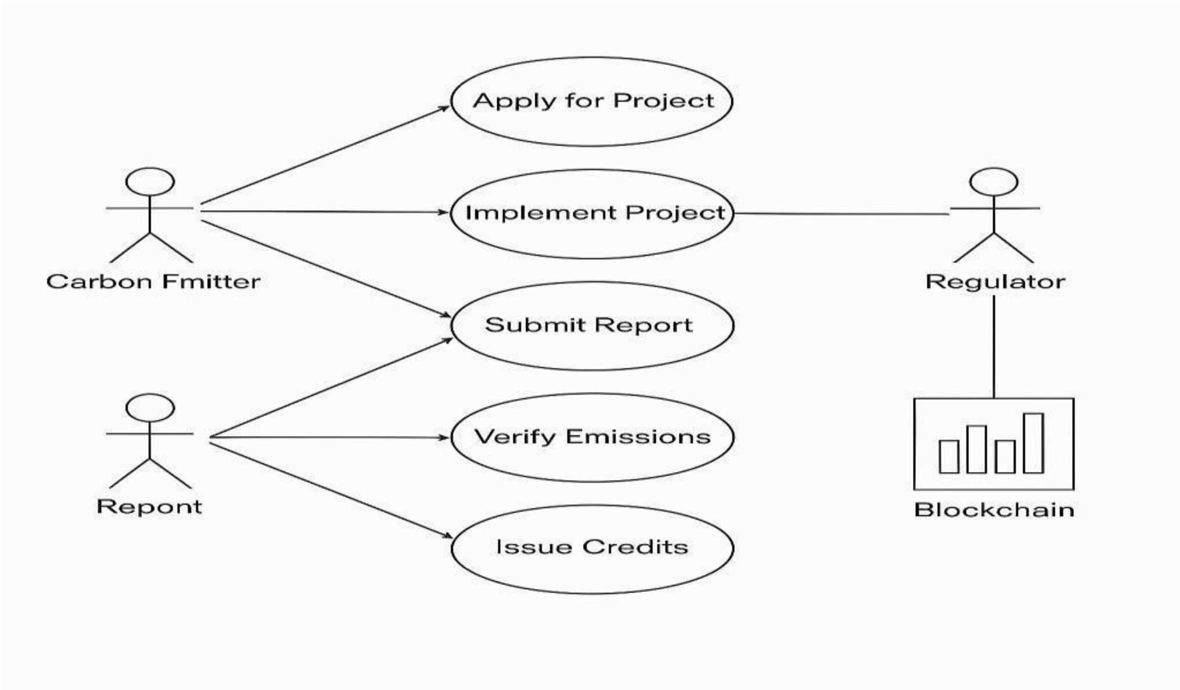

Figure 4.2.2: Usecase Diagram

4.2.3 Sequence Diagram

The sequence diagram represents the step-by-step flow of interactions involved in the verification and credit issuance process within the Carbon on the Chain platform. The process begins when the Carbon Emitter submits an emission report to the Regulator**,** who then forwards the data to the Smart Contract for validation. The smart contract performs automatic checks, records the validation event on the blockchain, and ensures the submitted report is securely stored and tamper-proof.

Once the verification is confirmed, the smart contract notifies the Regulator, enabling it to issue the corresponding carbon credits. The Regulator then triggers the credit issuance function, after which the smart contract mints the required ERC-20 carbon credit tokens and transfers them to the emitter's blockchain wallet. This sequence highlights the automated workflow and clear communication between actors—Emitter, Regulator, and Smart Contract—ensuring transparency, immutability, and efficiency in the carbon credit lifecycle.

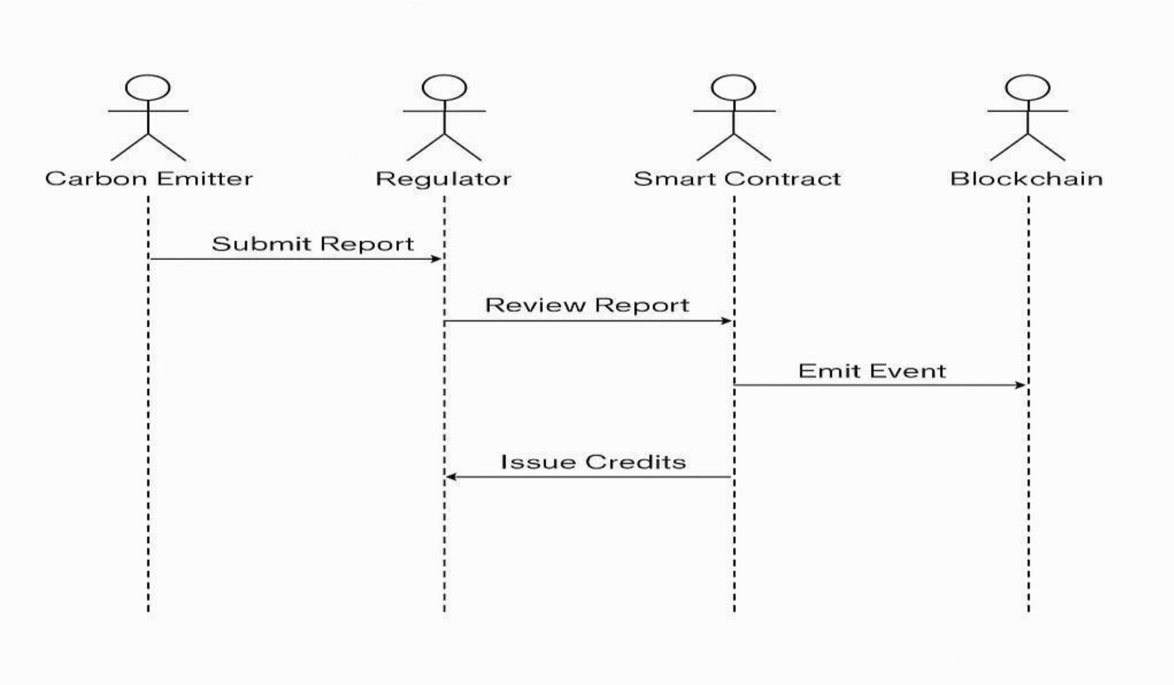

Figure 4.2.3: Sequence Diagram.

4.2.4 DataFlow Diagram

The Data Flow Diagram illustrates how information moves through the *Carbon on the Chain* platform and how each entity interacts with the system during the carbon credit lifecycle. The process begins with the Carbon Emitter, who inputs organization details and submits emission reports into the system.The submitted reports flow to the Verifier**,** who accesses the stored data to review and validate emission reduction claims.

The verification results are then passed to the Regulator, who evaluates the validated reports and initiates the issuance of carbon credits. Once approved, the system mints ERC-20 carbon credit tokens and transfers them to the emitter's wallet. All critical transactions—including submissions, validations, and credit issuance—are continuously recorded on the blockchain to ensure transparency and prevent tampering. The DFD thus represents a structured and secure flow of information between emitters, verifiers, regulators, and the blockchain system, ensuring an automated, trustworthy, and transparent carbon credit process.

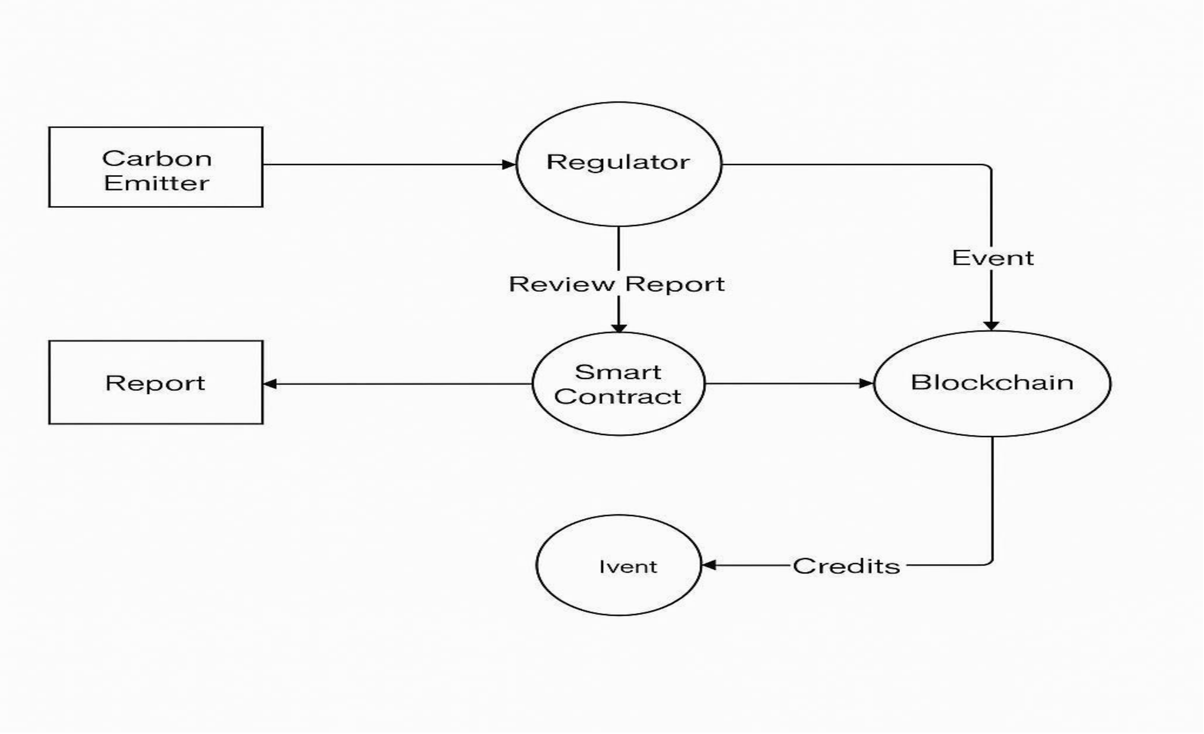

Figure 4.2.4: Data flow Diagram

CHAPTER 5

IMPLEMENTATION

5.1 Algorithms Used in Implementation

1. Registration and Authentication Algorithm

This algorithm verifies user identities (Emitter, Verifier, Regulator) based on their blockchain wallet addresses. When a user registers, the system checks the uniqueness of the address, validates mandatory fields, and stores the record on-chain. MetaMask functions as the authentication layer, ensuring secure, decentralized login without storing passwords.

2. Emission Submission Algorithm

This algorithm handles the intake of emission data from industries. It first validates the format and completeness of the emission report, uploads the supporting documents to IPFS, and stores the IPFS hash on the blockchain. This ensures immutability and tamper-proof recordkeeping.

3. Verification Algorithm (Auditor Validation)

This algorithm allows auditors to validate submitted emission reports. It ensures that only authorized verifiers can access specific reports, compares submitted values with expected standards, and updates the verification status on the blockchain.

- Authorization check
- Data integrity validation
- Approval or rejection
- Writing verification result to the smart contract

4. CAP Limit Evaluation Algorithm (BEE)

This algorithm compares the verified emission levels of an industry with the CAP limit set for its sector. If emissions are below the CAP, the industry qualifies for carbon credits.

Decision logic:

If (verifiedEmission < CAPLimit)
 credits = CAPLimit – verifiedEmission
Else
 credits = 0

5. Token Minting Algorithm (ERC-20 Carbon Credit Token)

Once emissions are verified and approved, the algorithm mints corresponding carbon credit tokens using the ERC-20 standard. It updates token balances, ensures supply control, and prevents unauthorized minting through role-based access.

1. Receive approved credit amount
2. Trigger mint() function
3. Update total supply & user balance
4. Record transaction hash on-chain

6. NFT Certificate Generation Algorithm (ERC-721)

This algorithm is used by POSOCO to issue digital certificates to registered industries. It generates a unique token ID, binds metadata with the organization's details, and creates an NFT representing industry approval.

7. Trading and Matching Algorithm (Carbon Credit Exchange)

This algorithm manages buy/sell orders of carbon credits. It records open orders, matches compatible buy–sell pairs based on price and quantity, and executes trades securely on the blockchain.

- Compare highest buy with lowest sell
- Execute trade when buy price ≥ sell price
- Update balances and remove completed orders

8. Blockchain Logging & Audit Trail Algorithm

Every critical action—registration, submission, verification, approval, minting, or trade—is logged on the blockchain. This ensures transparency, traceability, and prevents data tampering.

5.2 Modules

The system incorporates several advanced modules that work cohesively to deliver a decentralized and tamper-proof carbon governance framework. Each module is designed to automate critical operations, enforce trust through blockchain rules, and provide transparency to all stakeholders.

The Industry Registration Module enables organizations to register their industrial activities securely on the blockchain. This module captures essential details including industry type, registration ID, PAN/TAN, and ownership credentials. Once the data is validated by POSOCO, the system permanently stores the organization's identity in the CarbonRegistry smart contract, ensuring future traceability and preventing unauthorized participation.The POSOCO NFT Certification Module extends registration by issuing an ERC-721 POSNFT certificate to approved organizations. This non-fungible token acts as a decentralized digital identity, binding the organization's credentials to a cryptographically secure token. The NFT prevents identity fraud, ensures one-industry-one-certificate enforcement, and is used by the system for role-based access at every stage of the carbon lifecycle.

The Emission Report Submission Module allows industries to submit annual or periodic CO_2 emission readings through a Web3-enabled interface. Uploaded reports, sensor readings, and supporting documents are stored on IPFS to reduce cost and maintain data integrity. The system automatically hashes the file and links it to the blockchain registry, guaranteeing authenticity and preventing tampering or re-submission of the same report.

The Auditor Verification Module plays a crucial role in validating the submitted emission data. Accredited auditors evaluate the uploaded records, cross-check data consistency, compare against baseline values, and approve or reject the submission using a digital signature. Their decisions are recorded on-chain using the CarbonRegistry smart contract, enabling an immutable, publicly verifiable audit trail.

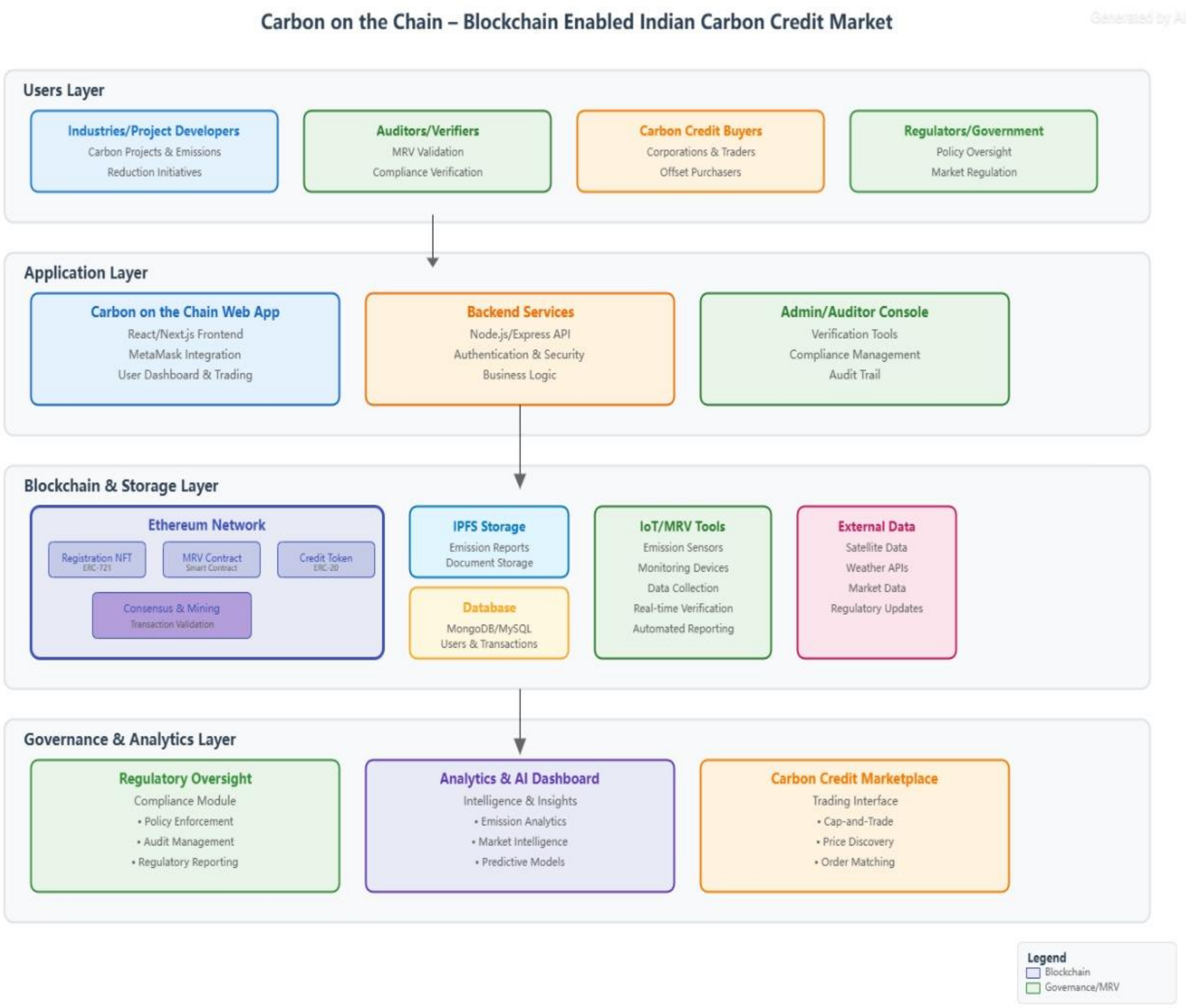

Figure 5.2.1: WorkFlow

The BEE Credit Reward Module processes verified data and calculates eligible carbon credits automatically. Once verification is confirmed, BEE interacts with the smart contract to assign reward points and authorize the minting of Carbon Credit Tokens (CCT). This module acts as the compliance authority, ensuring that industries are rewarded fairly according to their reduced emissions.

The ERC-20 Minting Module executes the automated creation of Carbon Credit Tokens through the CarbonCreditToken smart contract. The system calculates the number of credits to mint based on verified emission reductions and credits them directly to the organization's blockchain wallet. The module prevents double-minting, enforces credit scarcity, and maintains total supply logic using blockchain invariants.

The Token Trading Module enables decentralized exchange of CCT tokens between organizations, power producers, and traders. It supports buy/sell operations, escrow-based transactions, and price discovery through on-chain events. Since all transfers occur on the blockchain, traders receive complete transparency regarding token origin, ownership, and historical transactions.The PDF Certificate Generator Module creates downloadable compliance certificates for industries, based on auditor verification. Using jsPDF, the module generates two types of certificates—VALID and INVALID—each containing blockchain transaction IDs, verification timestamps, NFT details, and auditor remarks. These documents serve as formal proof for regulatory submissions and compliance audits.

Additionally, the Dashboard and Analytics Module integrates Chart.js to present emission trends, token balances, transaction history, and compliance statistics in an interactive, real-time format. This module empowers stakeholders with actionable insights, enabling better decision-making and system oversight.

Finally, the Blockchain Storage and Audit Trail Module ensures that every interaction—registration, NFT issuance, verification update, token minting, and trade—is permanently recorded on the blockchain. This module forms the backbone of the system’s trust mechanism by providing an immutable ledger that auditors, regulators, and industries can rely on for transparency and accountability.

CHAPTER 6

TESTING AND VALIDATION

6.1 Unit Testing and Test Cases

Unit testing was carried out for every individual module of the *Carbon on the Chain* system to ensure that each smart contract function, dashboard component, and data-processing workflow operated correctly before integration. Each module—such as Organization Registration, NFT Issuance, Emission Submission, Verification, BEE Reward Allocation, Token Minting, Certificate Generation, and Token Trading—was tested in isolation using sample inputs. Smart contract functions were tested on Ganache/Hardhat to verify correct state transitions, access control enforcement, event generation, and transaction success or failure.

Frontend units were tested by simulating user interactions such as submitting forms, connecting wallets, uploading documents, and triggering blockchain transactions. Edge cases, including invalid inputs, unauthorized access attempts, duplicate submissions, and insufficient balance scenarios, were also tested to evaluate error handling and system robustness. Through unit testing, each functional block of the system was validated for correctness, reliability, and stability before moving into integration and full-system testing.

Test Cases

1. Organization Registration

- Test registration with a new wallet address
- Test duplicate registration
- Test unauthorized role assignment (should fail)

2. NFT Certificate Issuance

- Issue NFT for approved organization
- Attempt duplicate issuance (should be blocked)
- Unauthorized NFT issue attempt

3. Emission Report Submission

- Valid CO_2 emission data + document upload
- Missing or malformed data (should give error)
- Invalid IPFS upload test

4. Auditor Verification

- Valid emission report approval
- Rejection of incorrect emission values
- Unauthorized verification access denial

5. BEE Credit Rewarding

- Verified emission $<$ CAP → credits issued
- Emission $\geq$ CAP → zero credits
- Invalid CAP values

6. ERC-20 Token Minting

- Mint tokens for verified credits
- Unauthorized mint attempt
- Mint with unrealistic values

7. Token Trading (Buy/Sell)

- Valid buy order placement
- Valid sell order placement
- Trade execution for matched orders
- Insufficient balance while trading (should fail)

8. Certificate Generation

- Generate VALID certificate after approval
- Generate INVALID certificate after rejection
- Export PDF successfully

6.2 Integration Testing and Test Cases

Integration testing was conducted to ensure that all modules of the *Carbon on the Chain* system work together seamlessly after unit testing. Since this platform involves multiple interconnected components—such as registration, NFT issuance, emission submission, auditor verification, CAP evaluation, credit rewards, token minting, certificate generation, and trading—each integration point was tested to confirm correct data flow and communication between smart contracts, dashboards, and blockchain storage.

The interactions between POSOCO → Organization → Auditor → BEE → Token Contract → Exchange were validated step-by-step to ensure that actions performed in one module correctly triggered the expected outcomes in the next. This included verifying that approved organizations received NFTs, verified emission reports triggered credit calculations, valid rewards resulted in token minting, and the minted tokens became usable for trading.

Integration testing also checked for error propagation, ensuring that invalid inputs, rejected verifications, or failed transactions did not break the workflow but returned appropriate feedback. Overall, integration testing confirmed that the entire blockchain-based system functioned cohesively as a unified application.

6.3 System Testing

System testing was carried out to validate the complete functionality, reliability, and performance of the blockchain-based carbon credit management system. The testing ensured that all integrated components—including registration, POSOCO NFT issuance, emission report submission, auditor verification, BEE credit allocation, ERC-20 token minting, and trading—worked together seamlessly as a unified application.

Unit tests were performed on each smart contract function to confirm accurate execution and error handling, while integration tests verified smooth data flow between the frontend dashboards, blockchain network, and off-chain storage. Functional testing validated that every user—POSOCO, BEE, Auditor, and Industry—could perform their designated activities without failure. Security testing ensured that unauthorized users could not bypass Metamask authentication, manipulate emission data, or access restricted functionalities.

Blockchain-specific testing validated immutability, correct event logging, token balance updates, gas usage, and prevention of double-spending. Performance testing demonstrated that contract execution, dashboard rendering, and token transfers occurred efficiently with minimal delays. Overall, the system passed all major test cases successfully, confirming that the application is secure, transparent, robust, and capable of supporting real-time carbon credit operations.

CHAPTER 7

RESULTS AND DISCUSSIONS

7.1 Snapshots

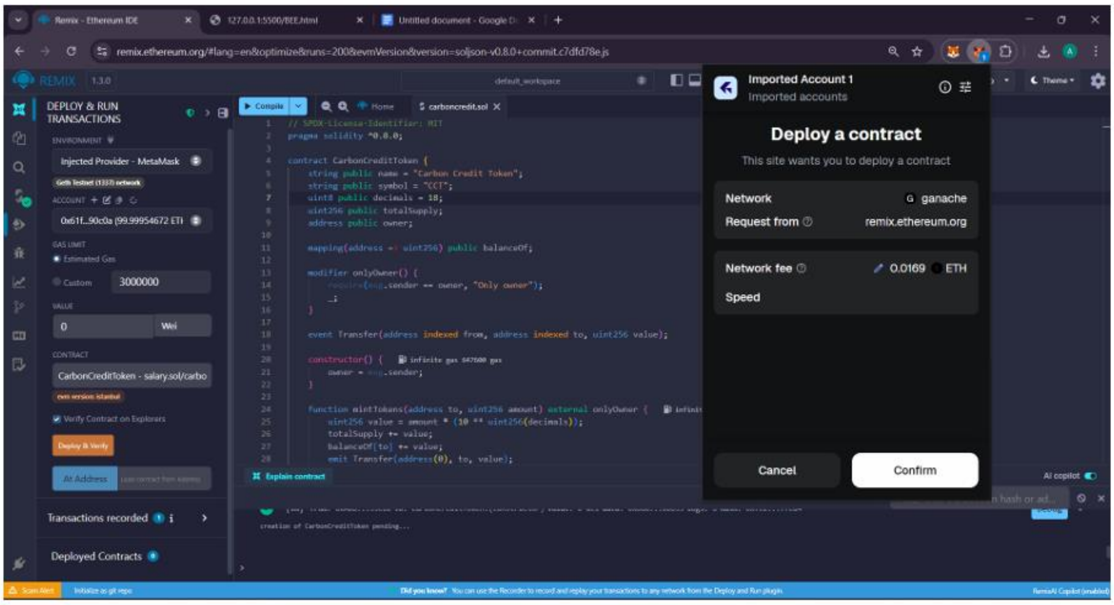

Figure 7.1.1: Remix connected with Metamask

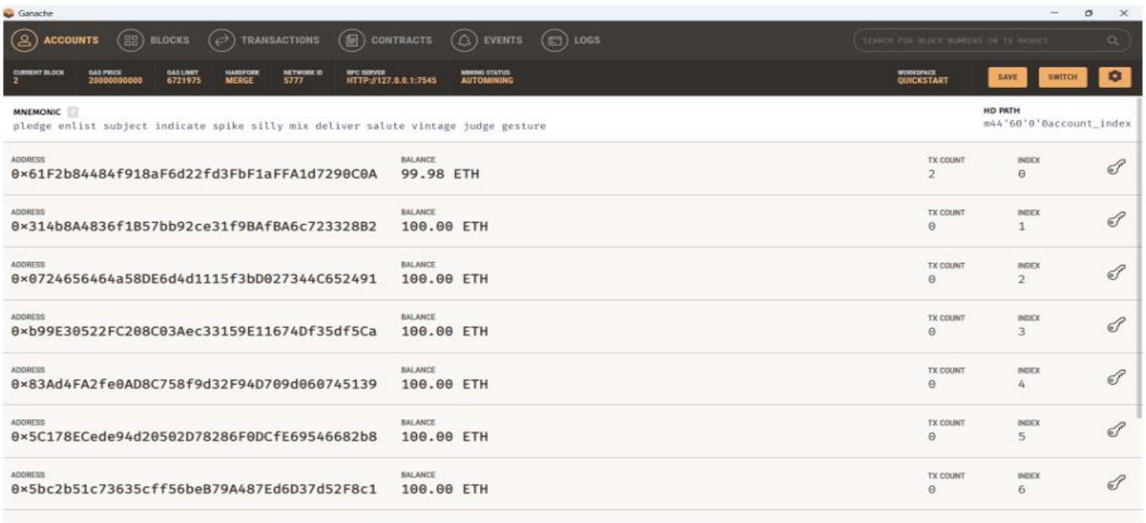

Figure 7.1.2: Ganache for wallet address.

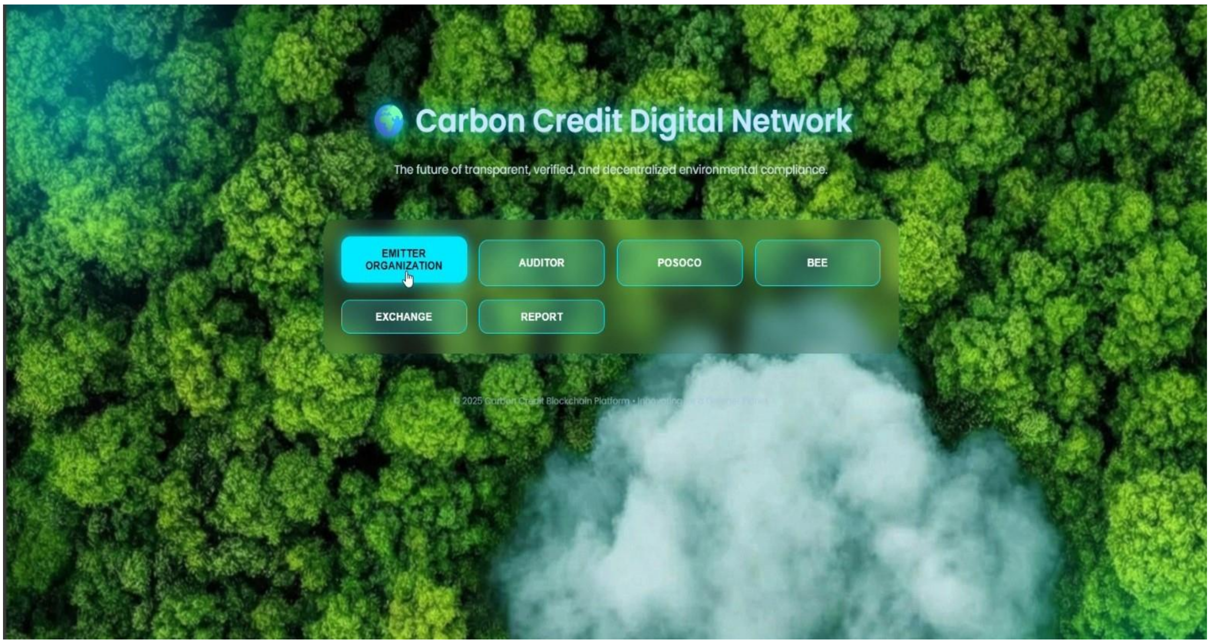

Figure 7.1.3: Dashboard for Carbon Credit.

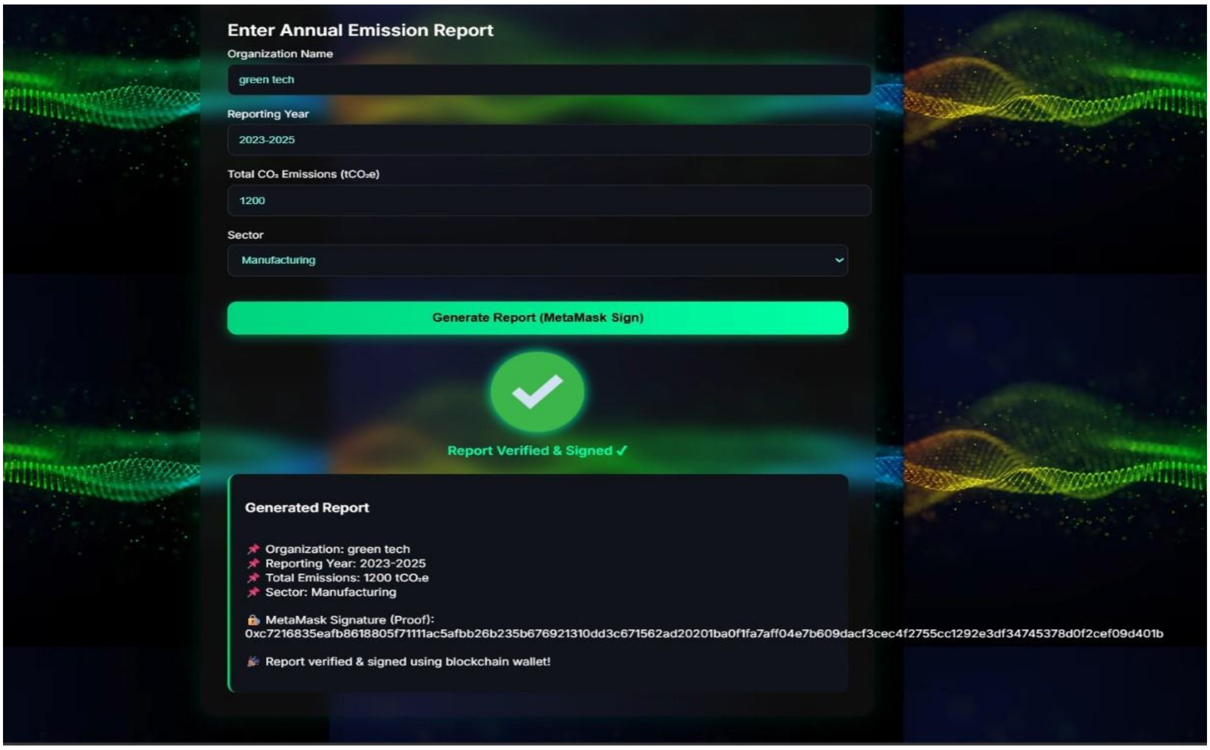

Figure 7.1.4: Emitter page for submitting Annual Report.

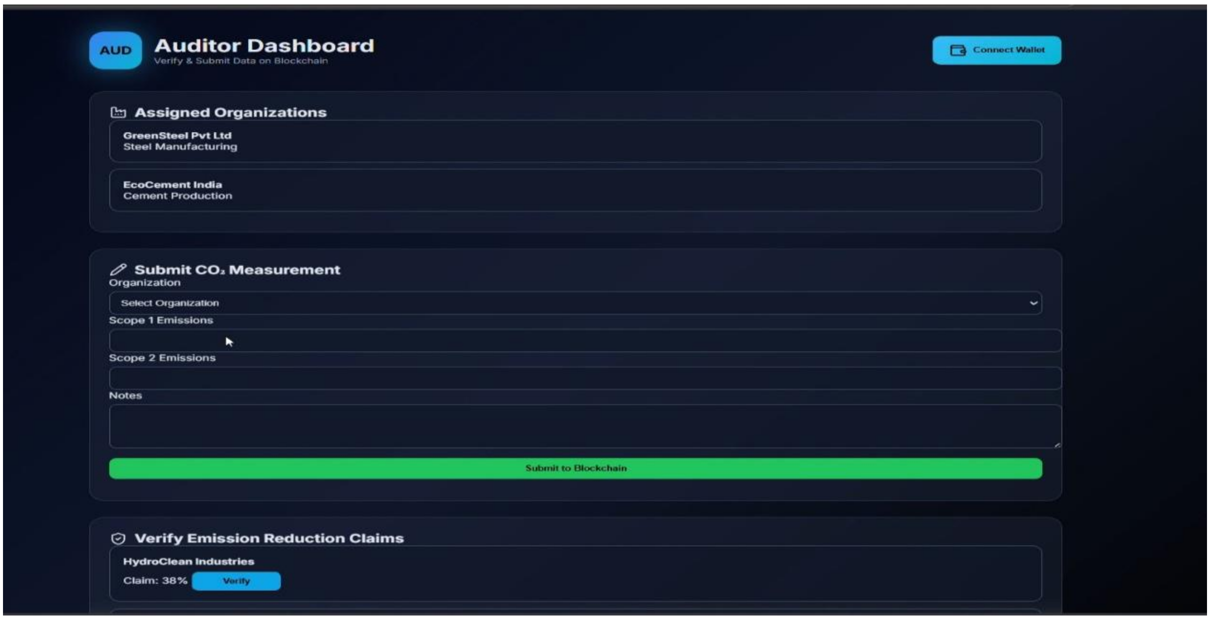

Figure 7.1.5: Auditor verifies Report Emisssions.

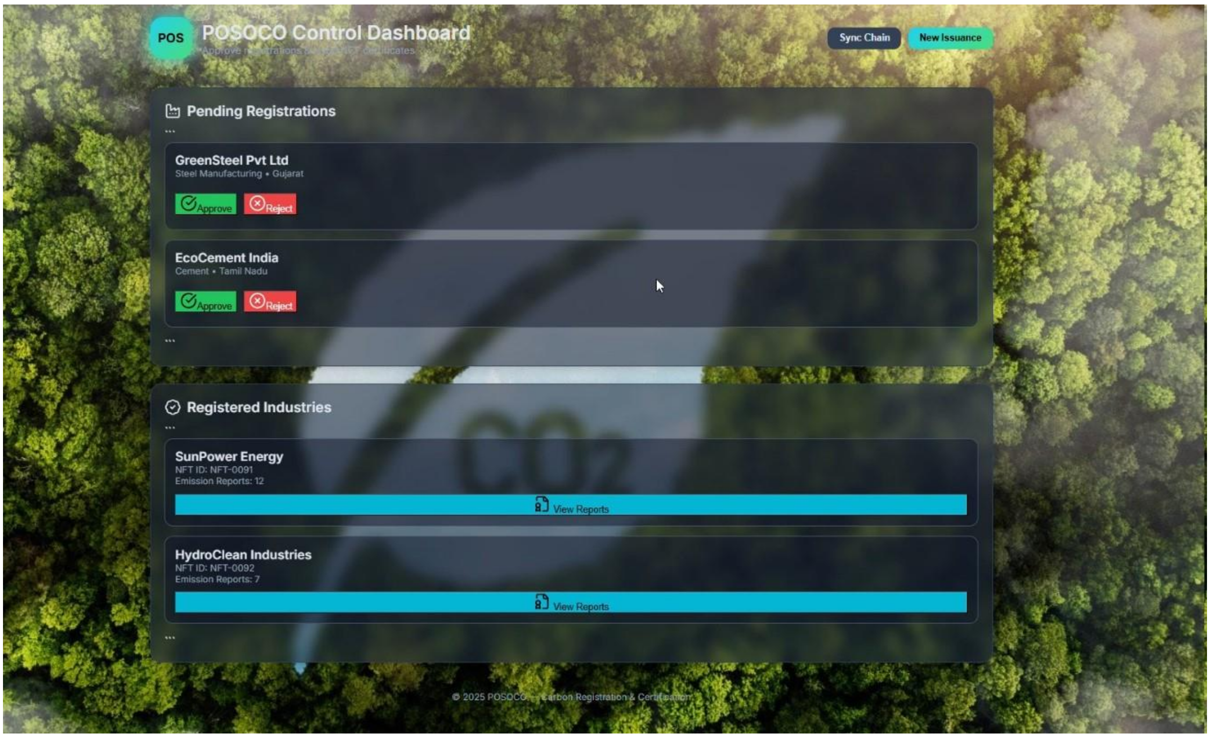

Figure 7.1.6: POSCO Dashboard for NFT tokens generating.

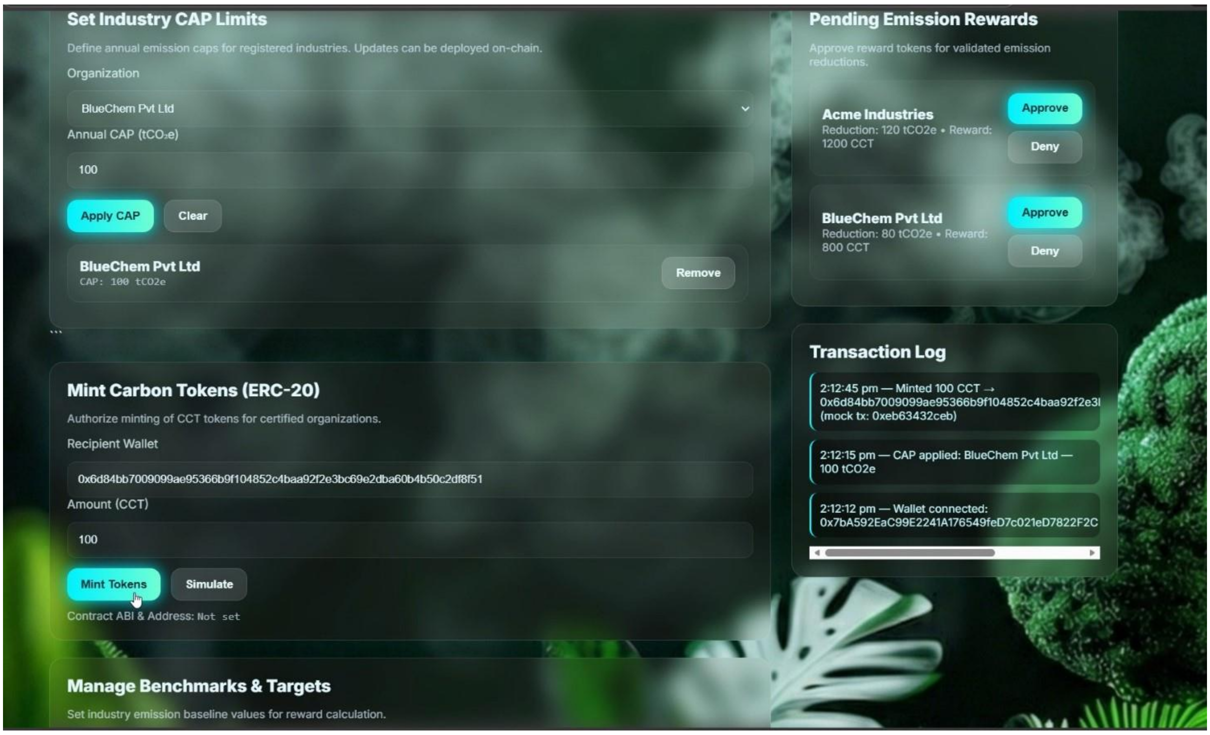

Figure 7.1.7: BEE to set CAP Limit and Minting Tokens.

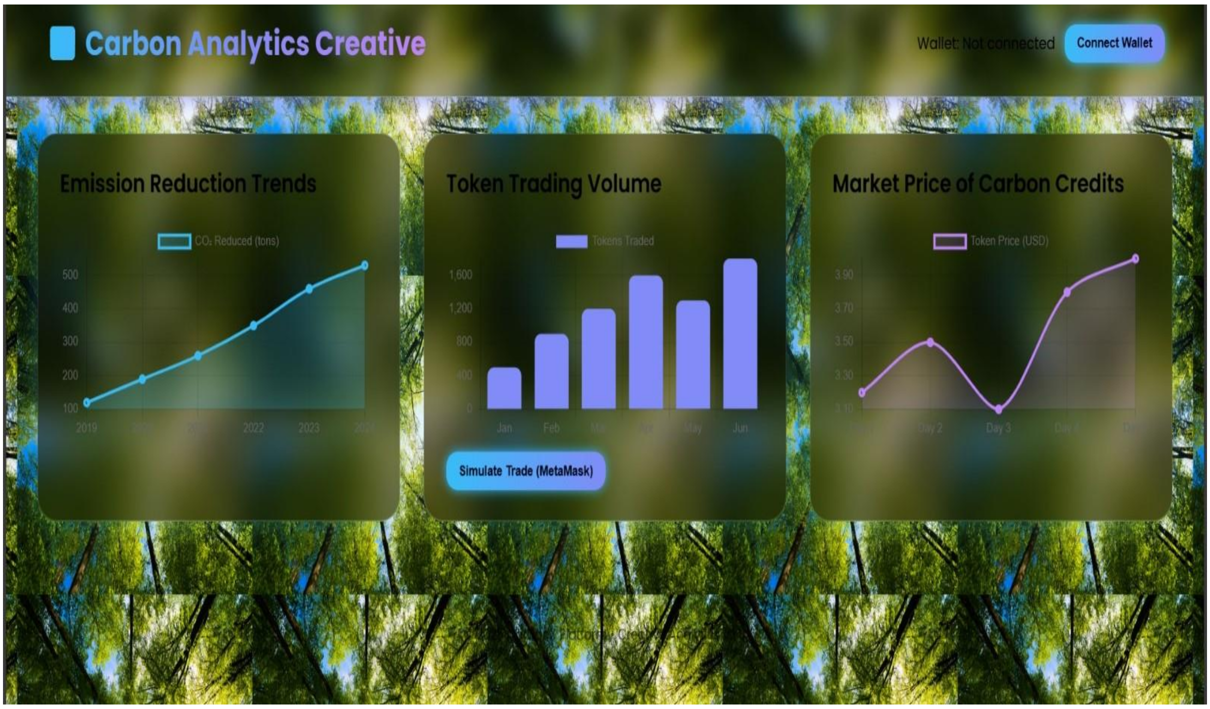

Figure 7.1.8: Overall Report of Carbon trading,reduction.

CONCLUSION

The project *"Carbon on the Chain: Digital Solutions for a Greener Planet"* successfully demonstrates how blockchain technology can modernize and streamline the carbon credit management process. By integrating smart contracts, decentralized verification, and tokenization, the system provides a transparent and tamper-proof framework for tracking emissions, issuing certificates, and managing carbon credits efficiently. The platform addresses major challenges in traditional carbon credit systems, such as data manipulation, manual verification delays, double-spending, and lack of real-time monitoring.

Through the implementation of ERC-721 NFTs for digital identity and ERC-20 tokens for carbon credits, the system ensures secure, traceable, and fraud-resistant transactions. Each stakeholder—POSOCO, BEE, auditors, and organizations—interacts through authenticated blockchain wallets, ensuring role-based access, trust, and accountability. The integration of IPFS storage further enhances security by preserving emission reports in an immutable and verifiable manner.

All key modules, including organization registration, emission submission, NFT certificate generation, auditor verification, credit reward minting, and token trading, were implemented and tested successfully. System testing confirmed the smooth functioning of both frontend and blockchain components and validated the robustness of smart contracts. Security, performance, and functional tests showed that the platform meets the required standards of reliability, accuracy, and data integrity.

The project highlights the potential of blockchain technology to support global climate governance by offering automated, transparent, and efficient carbon tracking solutions. This work can be extended with AI-driven analytics, IoT-based real-time emission monitoring, advanced trading mechanisms, and deployment on public blockchain networks. Overall, the system achieves its objectives effectively and contributes to creating a more sustainable, accountable, and technology-driven carbon management ecosystem.

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Project Outcome

Project title: Carbon on the Chain: Digital Solutions for a Greener a Planet. **Year:2025-26**

Sl No:	Factors addressed through project	Applicable POs and PSOs	Justification
1	Blockchain-based Carbon Credit Management	**PO1, PO2, PO3, PO5 & PO 6 PSO1**	Applied engineering knowledge to design and develop a secure, transparent, and tamper-proof carbon credit tracking system using modern tools and smart contracts. Demonstrated ability to solve real-world sustainability problems, ensured data integrity, and enhanced teamwork through collaborative development and documentation.

Guide Name & signature

Mrs. Kusuma C
Assistant Professor
Dept. of ISE

Students Name & Signature

Chandana S(1JB22IS032)
Deekshitha R(1JB22IS045)
Divya Devaraj Hegde(1JB22IS055)
Ganashree B S(1JB22IS062)